

```
clear; clc; close all;
```

```
%% LOAD & CLEAN DATASET
```

```
filePath = "C:\Users\Nishanth\OneDrive\Desktop\city_day.csv";  
T = readtable(filePath);
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property. Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
T.Properties.VariableNames = matlab.lang.makeValidName(T.Properties.VariableNames);  
fprintf('\nDataset loaded: %d rows, %d columns\n', height(T), width(T));
```

Dataset loaded: 29531 rows, 16 columns

```
fprintf('Cities found : %d\n\n', numel(unique(T.City)));
```

Cities found : 26

```
%% USER SELECTS PRIMARY POLLUTANT
```

```
fprintf('Available pollutants: PM2_5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3,  
Benzene, Toluene, Xylene\n');
```

Available pollutants: PM2_5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene

```
mainPollutant = input('Enter pollutant name for clustering (e.g. PM2_5): ', 's');  
  
if ~ismember(mainPollutant, T.Properties.VariableNames)  
    error("Pollutant '%s' not found in dataset!", mainPollutant);  
end
```

```
%% COMPUTE CITY-LEVEL FEATURES
```

```
% Features per city: Mean, Std, Median, 90th Percentile, Mean AQI
```

```
cities    = unique(T.City);  
numCities = numel(cities);  
  
featMean   = zeros(numCities, 1);  
featStd    = zeros(numCities, 1);  
featMedian = zeros(numCities, 1);  
featP90    = zeros(numCities, 1);  
featAQI    = zeros(numCities, 1);
```

```
for i = 1:numCities  
    mask = strcmp(T.City, cities{i});  
    vals = T.(mainPollutant)(mask);  
    aqi  = T.AQI(mask);
```

```

% Clean pollutant values
vals = vals(~isnan(vals));
vals = rmoutliers(vals, 'mean', 'ThresholdFactor', 3);

% Clean AQI values
aqi = aqi(~isnan(aqi));
aqi = rmoutliers(aqi, 'mean', 'ThresholdFactor', 3);

if numel(vals) < 10
    featMean(i) = NaN; featStd(i) = NaN;
    featMedian(i) = NaN; featP90(i) = NaN;
else
    featMean(i) = mean(vals);
    featStd(i) = std(vals);
    featMedian(i) = median(vals);
    featP90(i) = prctile(vals, 90);
end

if numel(aqi) < 5
    featAQI(i) = NaN;
else
    featAQI(i) = mean(aqi);
end
end

```

```

% Remove cities with insufficient data
valid = ~any([isnan(featMean) isnan(featStd) isnan(featMedian) ...
             isnan(featP90) isnan(featAQI)], 2);
featMean = featMean(valid);
featStd = featStd(valid);
featMedian = featMedian(valid);
featP90 = featP90(valid);
featAQI = featAQI(valid);
cities = cities(valid);

fprintf('Valid cities for clustering: %d\n\n', numel(cities));

```

Valid cities for clustering: 26

```

%% STEP 4 – NORMALISE FEATURE MATRIX
X = [featMean, featStd, featMedian, featP90, featAQI];
X_norm = zscore(X);

```

```

%% STEP 5 – FIND OPTIMAL K via SILHOUETTE SCORE

```

```
fprintf('Computing silhouette scores for k = 2 to 6 ...\n');
```

Computing silhouette scores for k = 2 to 6 ...

```
maxK      = min(6, numel(cities) - 1);  
silScores = zeros(maxK - 1, 1);  
  
for k = 2:maxK  
    [idxTmp, ~] = kmeans(X_norm, k, 'Replicates', 50, 'MaxIter', 500);  
    silScores(k-1) = mean(silhouette(X_norm, idxTmp));  
    fprintf('  k = %d | Silhouette = %.4f\n', k, silScores(k-1));  
end
```

```
k = 2 | Silhouette = 0.5894  
k = 3 | Silhouette = 0.6583  
k = 4 | Silhouette = 0.6318  
k = 5 | Silhouette = 0.5664  
k = 6 | Silhouette = 0.5966
```

```
[~, bestIdx] = max(silScores);  
kOpt = bestIdx + 1;  
fprintf('\n>> Optimal k = %d\n\n', kOpt);
```

```
>> Optimal k = 3
```

```
%% STEP 6 – FINAL K-MEANS CLUSTERING  
rng(12);  
[idx, ~] = kmeans(X_norm, kOpt, 'Replicates', 100, 'MaxIter', 1000, ...  
                  'Distance', 'sqeuclidean');
```

```
%% STEP 7 – LABEL CLUSTERS BY POLLUTION LEVEL  
clusterMeanVal = zeros(kOpt, 1);  
for c = 1:kOpt  
    clusterMeanVal(c) = mean(featMean(idx == c));  
end  
[~, order] = sort(clusterMeanVal);  
levelNames = ["Low Pollution", "Moderate Pollution", "High Pollution", ...  
              "Very High Pollution", "Extreme Pollution", "Hazardous Pollution"];  
clusterLabels = strings(kOpt, 1);  
for c = 1:kOpt  
    clusterLabels(order(c)) = levelNames(c);  
end
```

```
%% COLOUR SCHEME
```

```
colors = [0.18 0.55 0.34; % green – Low
```

```

0.95 0.77 0.06; % yellow – Moderate
0.85 0.33 0.10; % orange – High
0.64 0.08 0.18; % red – Very High
0.30 0.00 0.36; % purple – Extreme
0.05 0.05 0.05]; % black – Hazardous
colors = colors(1:kOpt, :);

```

```

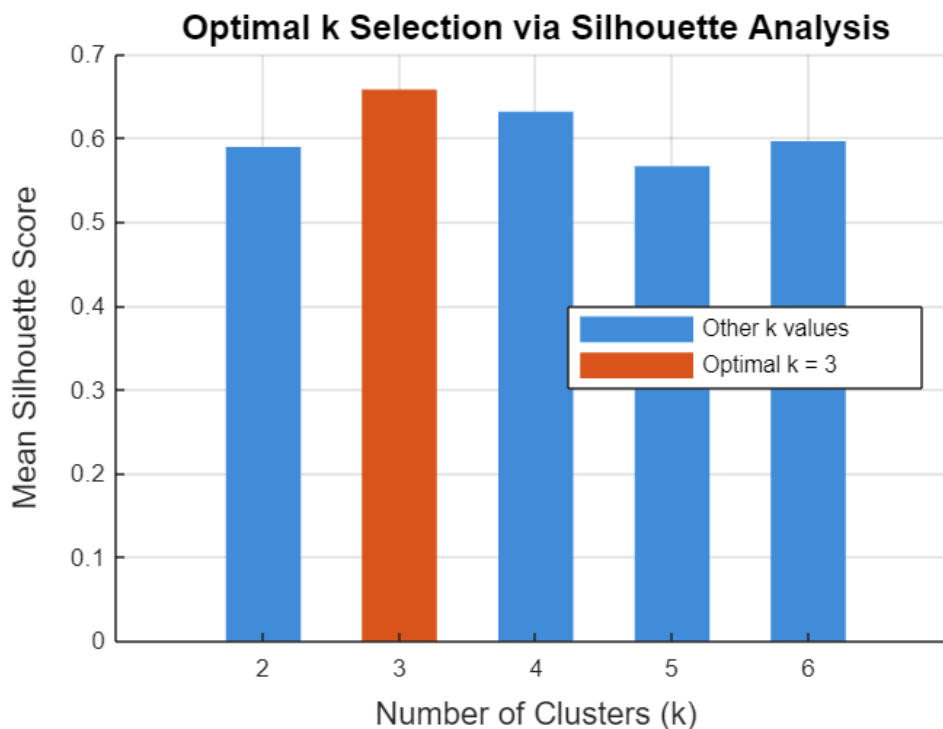
%% FIGURE 1 – Silhouette Bar Chart (Optimal k)

```

```

figure('Name', 'Silhouette Analysis', 'Color', 'w', 'Position', [100 100 540 380]);
bar(2:maxK, silScores, 0.55, 'FaceColor', [0.25 0.55 0.85], 'EdgeColor', 'none');
hold on;
bar(kOpt, silScores(kOpt-1), 0.55, 'FaceColor', [0.85 0.33 0.10], 'EdgeColor',
'none');
xlabel('Number of Clusters (k)', 'FontSize', 12);
ylabel('Mean Silhouette Score', 'FontSize', 12);
title('Optimal k Selection via Silhouette Analysis', 'FontSize', 13, 'FontWeight',
'bold');
xticks(2:maxK);
legend({'Other k values', sprintf('Optimal k = %d', kOpt)}, 'Location', 'best');
grid on; box off;

```



```

%% FIGURE 2 – Scatter: Mean vs Std Dev

```

```

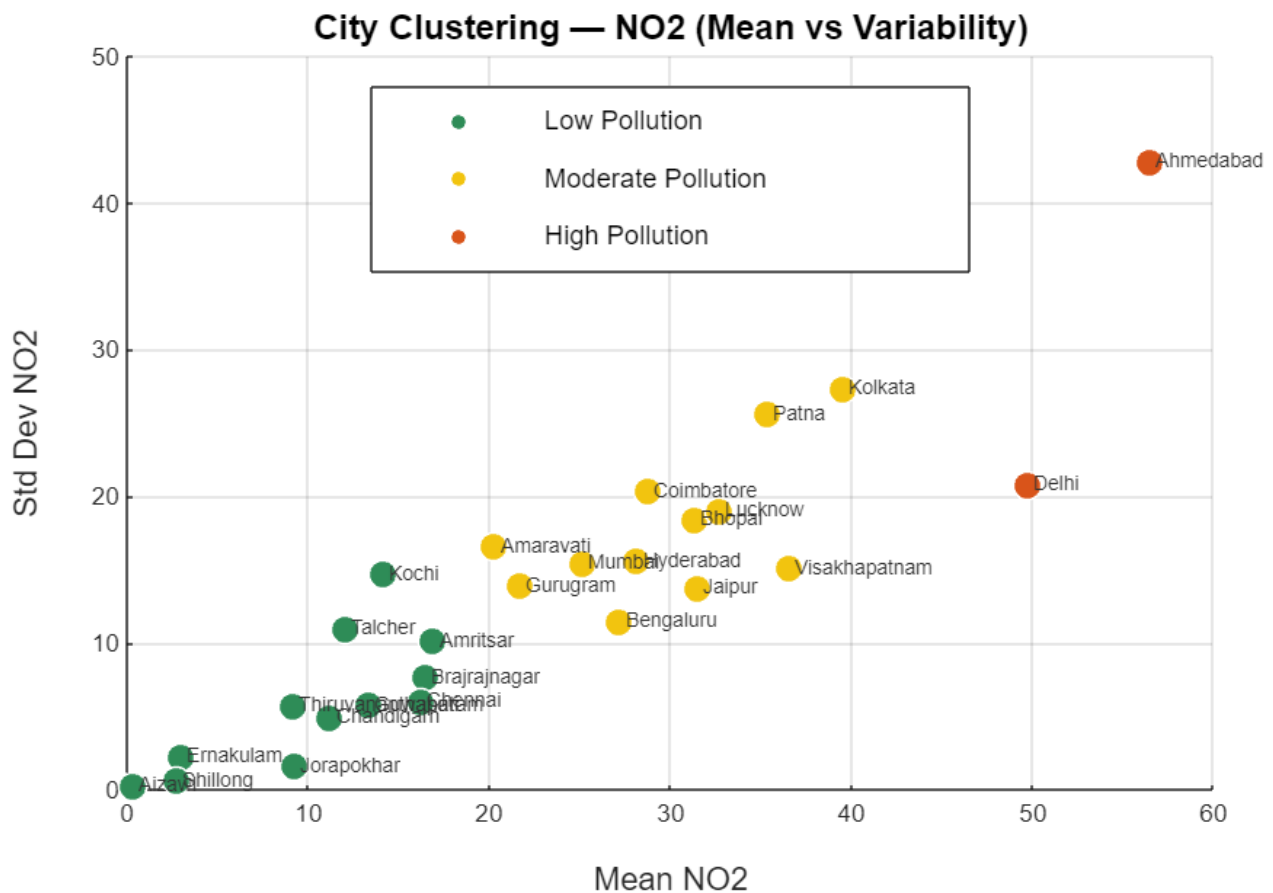
figure('Name', 'Clustering: Mean vs Std', 'Color', 'w', 'Position', [660 100 700
480]);

```

```

hold on;
for c = 1:kOpt
    mask = idx == c;
    scatter(featsMean(mask), featsStd(mask), 110, colors(c,:), 'filled', ...
        'MarkerEdgeColor', 'w', 'LineWidth', 0.8, 'DisplayName',
clusterLabels(c));
end
for i = 1:numel(cities)
    text(featsMean(i) + 0.3, featsStd(i), cities{i}, 'FontSize', 7.5, 'Color', [0.2
0.2 0.2]);
end
xlabel(['Mean ' mainPollutant], 'FontSize', 12);
ylabel(['Std Dev ' mainPollutant], 'FontSize', 12);
title(['City Clustering — ' mainPollutant ' (Mean vs Variability)'], ...
    'FontSize', 13, 'FontWeight', 'bold');
legend('Location', 'best', 'FontSize', 10);
grid on; box off;

```



% FIGURE 3 – Horizontal Bar: Cities Ranked by Mean Pollutant

```

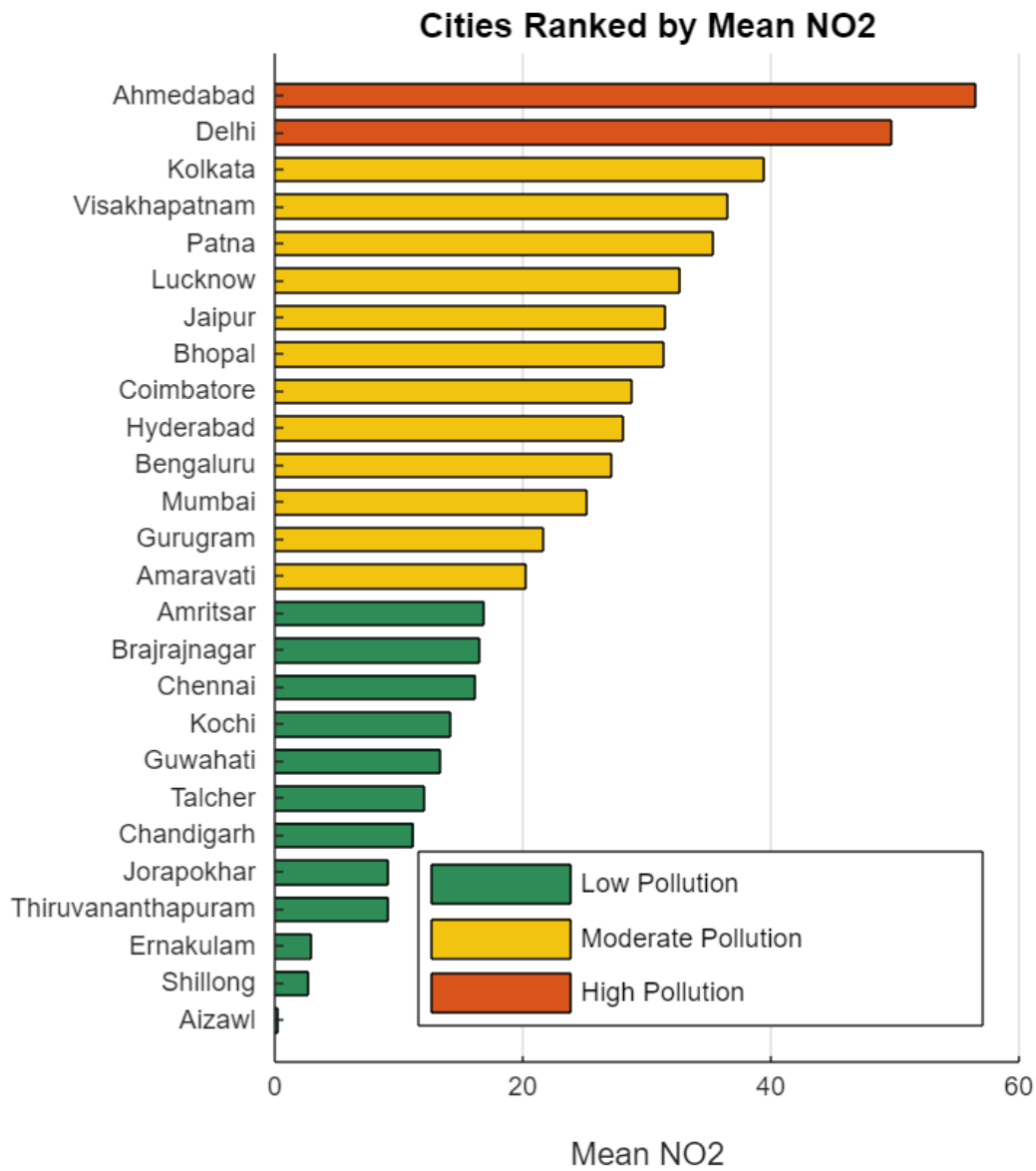
figure('Name', 'City Ranking', 'Color', 'w', 'Position', [820 540 660 620]);
[sortedMean, sortOrd] = sort(featsMean);

```

```

sortedCities = cities(sortOrd);
barColors     = colors(idx(sortOrd), :);
bh = barh(1:numel(cities), sortedMean, 0.65, 'FaceColor', 'flat');
bh.Annotation.LegendInformation.IconDisplayStyle = 'off';
bh.CData = barColors;
yticks(1:numel(cities));
yticklabels(sortedCities);
xlabel(['Mean ' mainPollutant], 'FontSize', 12);
title(['Cities Ranked by Mean ' mainPollutant], 'FontSize', 13, 'FontWeight',
'bold');
hold on;
for c = 1:kOpt
    patch(NaN, NaN, colors(c,:), 'DisplayName', clusterLabels(c));
end
legend('Location', 'southeast', 'FontSize', 10);
grid on; box off;
ax = gca; ax.XGrid = 'on'; ax.YGrid = 'off';

```



%% RESULT TABLE

```
silVals = silhouette(X_norm, idx);

ResultTable = table(cities, round(featsMean,2), round(featsStd,2), ...
                    round(featsMedian,2), round(featsP90,2), ...
                    round(featsAQI,1), clusterLabels(idx), round(silVals,3), ...
                    'VariableNames',
                    {'City','Mean','Std','Median','P90','Mean_AQI','Cluster','Silhouette'});

ResultTable = sortrows(ResultTable, 'Mean', 'descend');
```

```
fprintf('\n===== CITY CLUSTERING RESULTS (%s) =====\n', mainPollutant);
```

```
===== CITY CLUSTERING RESULTS (N02) =====
```

```
disp(ResultTable);
```

City	Mean	Std	Median	P90	Mean_AQI	Cluster	Silhouette
{ 'Ahmedabad' }	56.49	42.79	45.91	116.59	428.9	"High Pollution"	0.627
{ 'Delhi' }	49.72	20.8	46.71	78.54	258.2	"High Pollution"	-0.26
{ 'Kolkata' }	39.51	27.35	32	81.71	139.7	"Moderate Pollution"	0.654
{ 'Visakhapatnam' }	36.57	15.07	35.23	57.41	111.8	"Moderate Pollution"	0.8
{ 'Patna' }	35.38	25.62	27.21	73.29	240.5	"Moderate Pollution"	0.539
{ 'Lucknow' }	32.73	19.02	28.05	60.3	216.7	"Moderate Pollution"	0.79
{ 'Jaipur' }	31.5	13.73	29.65	49.87	131.3	"Moderate Pollution"	0.79
{ 'Bhopal' }	31.35	18.41	32.5	56.3	130.3	"Moderate Pollution"	0.843
{ 'Coimbatore' }	28.78	20.33	23.99	60.29	73	"Moderate Pollution"	0.751
{ 'Hyderabad' }	28.16	15.57	25.55	49.49	105.6	"Moderate Pollution"	0.749
{ 'Bengaluru' }	27.2	11.48	26.81	42.38	91.6	"Moderate Pollution"	0.551
{ 'Mumbai' }	25.18	15.42	21.28	47.64	103.8	"Moderate Pollution"	0.611
{ 'Gurugram' }	21.69	13.91	19.17	41.29	222	"Moderate Pollution"	0.344
{ 'Amaravati' }	20.29	16.59	14.9	44.97	91.5	"Moderate Pollution"	0.147
{ 'Amritsar' }	16.91	10.12	13.51	31.67	115.9	"Low Pollution"	0.602
{ 'Brajrajnagar' }	16.49	7.64	16.06	26.98	150.3	"Low Pollution"	0.608
{ 'Chennai' }	16.23	6.01	15.34	23.86	111.1	"Low Pollution"	0.75
{ 'Kochi' }	14.15	14.67	9.26	29.84	102.2	"Low Pollution"	0.606
{ 'Guwahati' }	13.37	5.75	11.94	21.66	136.3	"Low Pollution"	0.825
{ 'Talcher' }	12.05	10.97	8.44	27.87	170.4	"Low Pollution"	0.678
{ 'Chandigarh' }	11.21	4.93	11.23	17.81	90.4	"Low Pollution"	0.876
{ 'Jorapokhar' }	9.21	1.61	9.36	11.22	151.7	"Low Pollution"	0.851
{ 'Thiruvananthapuram' }	9.18	5.69	7.55	17.61	73.7	"Low Pollution"	0.881
{ 'Ernakulam' }	2.99	2.23	2.39	6.99	90.5	"Low Pollution"	0.867
{ 'Shillong' }	2.72	0.59	2.67	3.55	51.2	"Low Pollution"	0.835
{ 'Aizawl' }	0.33	0.28	0.26	0.7	34.8	"Low Pollution"	0.803

```
fprintf('\n--- Cluster Summary ---\n');
```

```
--- Cluster Summary ---
```

```
for c = 1:kOpt
    mask = strcmp(ResultTable.Cluster, clusterLabels(c));
    memberCities = strjoin(ResultTable.City(mask), ', ');
    fprintf('%s (%d cities): %s\n', clusterLabels(c), sum(mask), memberCities);
end
```

```
Low Pollution (12 cities): Amritsar, Brajrajnagar, Chennai, Kochi, Guwahati, Talcher, Chandigarh, Jorapokhar, Thiruvananthapuram, Ernakulam, Shillong, Aizawl
Moderate Pollution (12 cities): Kolkata, Visakhapatnam, Patna, Lucknow, Jaipur, Bhopal, Coimbatore, Hyderabad, Bengaluru, Mumbai, Gurugram, Amaravati
High Pollution (2 cities): Ahmedabad, Delhi
```

```
fprintf('\nOverall mean silhouette score: %.4f\n', mean(silVals));
```

```
Overall mean silhouette score: 0.6583
```